

Leo Shaw

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Education

Carnegie Mellon University

Master of Science in Artificial Intelligence Engineering, Mechanical Engineering, GPA: 3.67/4.0

Pittsburgh, PA

December 2026

Relevant Courses: Modern Control Theory, Robot Dynamics and Analysis, Systems and Toolchains

Rutgers University

Bachelor of Science in Mechanical Engineering; Bachelor of Arts in Computer Science

New Brunswick, NJ

December 2024

Honors: Dean's List (2021-2022), Rutgers Badminton Club Division [D1] (President/Captain, 2021-2024)

Skills

Design/Analysis: SOLIDWORKS, ANSYS, MATLAB

Programming/AI: Python, C++, Java, PyTorch, TensorFlow, Keras, OpenCV, YOLO

Hardware/Fabrication: Mills, lathes, band saws, circuitry, 3D printers, soldering

Experience

Field Service Engineer I

Santa Clara, CA / Ulvac Technologies, October 2024 - August 2025

- Rebuilt and installed an upgraded 8-inch etch chuck assembly, integrated a new VAT system and circuit breaker, and modified valve-control software and wiring, improving tool flexibility, reducing unplanned downtime by 15%, and increasing wafer temperature uniformity.
- Maintained semiconductor tools across 6 high-volume fabs through preventive maintenance, calibration, and data-driven repairs, increasing uptime by 18%.
- Supported Intel, Apple, Google, and Penn State Nanofabrication Facility by resolving high-impact tool issues under tight production constraints.
- Executed 30+ hardware and software retrofit projects across vacuum, RF, motion-handling, PLC, and HMI subsystems, improving process stability by 22% and extending tool life.
- Led root-cause analysis with engineering and manufacturing teams, implemented corrective actions, and standardized troubleshooting procedures, reducing average fault diagnosis time by 35%.
- Authored 10+ technical reports, procedures, and maintenance checklists that improved technician training and reduced repeat failures.

Mechanical Engineering Intern

Milpitas, CA / Paltorc, May 2022 - August 2022

- Integrated the E-Bike control module with a mobile app to stream heart rate, temperature, weather, and incline data in real time.
- Diagnosed and resolved mechanical and software issues to improve system reliability.

Projects

Solar-Powered Terrain Walker / August 2023 - May 2024

- Designed and built an autonomous solar-powered quadruped using Chebyshev 4-bar linkages and pantographs, reducing cost by 34% versus comparable 6-legged designs.
- Reduced leg actuation from 3 motors to 1 through kinematic linkage design, improving energy efficiency and simplifying control.
- Modeled pivot mechanisms in SOLIDWORKS, validated structural integrity in ANSYS, and implemented motion and sensing subsystems in C on Arduino.

Autonomous Driving Local Planner with Reinforcement Learning / January 2026 - Present

- Built a hierarchical autonomous-driving planner in CARLA where a global A* route provides long-horizon guidance and a PPO policy generates reactive local trajectories for highway merging and unprotected left turns.
- Developed the training and evaluation pipeline in Python using Stable-Baselines3, PyTorch, and CARLA, including scenario generation, traffic interaction features, reward design, and rollout analysis tools.
- Achieved a 92% success rate using PPO and benchmarked performance against rule-based CARLA baselines using completion time, jerk-based comfort metrics, and failure-mode analysis.

Real-Time Conveyor-Belt Object Detection and Robotic Sorting / January 2026 - Present

- Developed a real-time computer vision pipeline in Python using YOLO and OpenCV to detect bottles, cans, and six-pack bundles on a conveyor belt.
- Improved object localization by refining centroids from object silhouettes and projecting detections into world coordinates with homography-based calibration.
- Reduced misclassification through confidence-weighted tracking, label stabilization across frames, and rapid auto-labeling and retraining.
- Programmed, assembled, tested, and integrated a delta-arm robotic structure for downstream sorting and material handling.